

SYMMETRISED MULTIVARIATE DISTRIBUTIONS

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Key words: symmetrised distributions, characteristic roots, Noncentral Beta densities, generalised multiple correlation matrix

1. Introduction

In multivariate analysis there exists a large class of important hypothesis testing problems all of which may be tested by a set of criteria that depend functionally on a matrix variate with a Beta distribution. Examples of such hypotheses are the following.

- (i) Hypothesis of independence of two sets of variates, considering the second set fixed. (We shall deal later with the case when both sets are comprised of random variables.)
- (ii) Linear hypotheses about regression coefficients.
- (iii) General linear hypothesis in MANOVA.

These may be tested using, for example, one of the following test criteria.

- (i) Wilks' likelihood ratio criterion (Wilks (1932), Anderson (1958)).
- (ii) Roy's largest and smallest root criteria (Roy (1957)).
- (iii) Hotelling's T^2_0 or Pillai's $U^{(P)}$ criterion (Pillai (1955)).
- (iv) Pillai's V and Q criteria (Pillai (1955)).

More specifically, these criteria are all functions of the characteristic roots of a p-dimensional $M\beta_{1b}(m,n,\Omega)$ matrix. The latter notation is explained in the following definition.

Definition 1.1

Let $A:pxp$ and $B:pxp$ be independently distributed according to the central Wishart distribution $W(\Sigma,m)$ and the noncentral Wishart distribution $W(\Sigma,n,\Omega)$ with non-centrality parameter matrix Ω respectively. Then

$$L = (A+B)^{-1/2}B(A+B)^{-1/2}$$

has the noncentral Beta type 1b* distribution with m and n degrees of freedom and noncentrality parameter matrix Ω , i.e. $L \sim M\beta_{1b}(m,n,\Omega)$.

* cf. the multivariate Beta type 1a distribution (3.3). The distinction is only necessary in the noncentral case where A and B differ in their noncentralities.

A first step in the investigation of the non-null distributions of these criteria is thus to derive the noncentral $M\beta_{1b}$ density function and the joint density of the roots of a noncentral $M\beta_{1b}$ matrix. The purpose of this note is to introduce the notion of a "symmetrised distribution" of a matrix variate, a notion which facilitates many proofs in such derivations. This new class of distributions will be seen to have very desirable properties in this context where it is in certain senses equivalent to the actual distribution and may in fact be used in its place. Apart from computational advantages the following further advantage of symmetrisation is apparent: whereas the form of the actual $M\beta_{1b}$ density is an unsolved integral, the symmetrised equivalent is solved and is in the very pleasing form of the central density multiplied by a weighting function. Similar results will be illustrated for other multivariate Beta densities and the density of the generalised multiple correlation matrix.

2. Definition and properties

The notion of a symmetrised density of a matrix variate is developed mainly from two papers by James (1955, 1960). Here he is concerned with the central Wishart distribution for general parameter matrix Σ and the distribution of the roots of such a Wishart matrix variate. In the latter paper he uses the term "symmetrising the density" to mean replacing the function

$$(2.1) \quad \text{etr}(-\frac{1}{2}\Sigma^{-1}XX')$$

by its average

$$(2.2) \quad \int_{O(p)} \text{etr}(-\frac{1}{2}\Sigma^{-1}HXX'H') d(H)$$

with respect to the invariant measure $d(H)$ on the orthogonal group $O(p)$ of $p \times p$ orthogonal matrices H . The invariant measure is normalised to make the total measure of $O(p)$ unity. The function (2.1) is not invariant under congruence transformation by the orthogonal group $O(p)$,

$$XX' \rightarrow HXX'H', \quad H \in O(p)$$

whereas the symmetrised function (2.2), which by James (1960) and Constantine (1963) is equal to

$${}_0F_0(-\frac{1}{2}\Sigma^{-1}, XX')$$

is now a symmetric function of the characteristic roots of Σ and the characteristic roots of XX' . The fact that the distribution of the roots of XX' is not altered if such a replacement is made in the density function justifies the argument.

The above ideas will now be extended and defined more specifically for matrix

density functions.

Definition 2.1

The symmetrised (multivariate) density of the positive definite symmetric matrix $A: p \times p$ is defined as

$$(2.3) \quad f_s(A) \equiv \int_{O(p)} f(HAH') d(H)$$

where A has the density function $f(A)$.

The symmetrised density clearly satisfies the conditions of a density function, and has the following important property.

Theorem 2.1

Let Φ be a function of the random matrix $A: p \times p$ such that

$$\Phi(A) = \Phi(H'AH)$$

for all orthogonal $p \times p$ matrices H . Then the distribution of $\Phi(A)$ is invariant with respect to symmetrisation of the density of A .

Proof: (A generalisation of the proof by James (1955))

Let M be a measurable set in the range of Φ and let $\Phi^{-1}(M)$ be its preimage. Let $\mu(M)$ and $\nu(M)$ be respective probabilities of M when A has the symmetrised and unsymmetrised distributions. Then

$$\mu(M) = \int_{\Phi^{-1}(M)} \left[\int_{O(p)} f(HAH') d(H) \right] dA$$

By changing the order of integration and transforming $A \rightarrow H'AH$, $H \in O(p)$, with Jacobian equal to $|H|^{p+1} = 1$ (Deemer & Olkin (1959), we obtain

$$\mu(M) = \int_{O(p)} d(H) \left[\int_{H'\Phi^{-1}(M)H} f(A) dA \right]$$

But $H'\Phi^{-1}(M)H = \Phi^{-1}(M)$, since $\Phi(H'AH) = \Phi(A)$. Therefore

$$\begin{aligned} \mu(M) &= \int_{O(p)} d(H) \int_{\Phi^{-1}(M)} f(A) dA \\ &= \int_{\Phi^{-1}(M)} f(A) dA \quad (\text{since } \int_{O(p)} d(H) = 1) \\ &= \nu(M) \end{aligned}$$

Q.E.D.

Thus using the symmetrised density of A the densities of all functions of A which are invariant under congruence transformations of A by the orthogonal group $O(p)$ can be obtained. Amongst these functions are $|A|$, trace A (i.e. $\sum_{i=1}^p a_{ii}$), in fact all functions which depend on A through its characteristic roots only. In addition the moments of these functions are invariant with respect to symmetrisation of the density of A .

To make use of symmetrised densities in the next section the following theorem, due to Anderson (1958, pp. 318-320), is required.

Theorem 2.2

If the symmetric matrix $S: p \times p$ is always nonsingular and if it has a density function which depends only on the characteristic roots of S , i.e.

$$(2.4) \quad f(S) = g(\lambda_1, \dots, \lambda_p),$$

where $\lambda_1 > \dots > \lambda_p > 0$ are the roots of S ,

then the joint distribution of the characteristic roots is

$$(2.5) \quad h(\lambda_1, \dots, \lambda_p) = \frac{\pi^{1/2} p^2}{\Gamma_p(1/2 p)} g(\lambda_1, \dots, \lambda_p) \prod_{i < j}^p (\lambda_i - \lambda_j).$$

3. Multivariate Beta densities

We are now in a position to return to Definition 1.1 and derive the relevant density functions.

Theorem 3.1

Let $L \sim M\beta_{1B}(m, n, \Omega)$; matrices A, B as before. Then

$$(3.1) \quad (i) \quad f(L) = \frac{1}{\Gamma_p(1/2 m) \Gamma_p(1/2 n) |2\Sigma|^{1/2(m+n)}} \text{etr}(-1/2 \Omega) |L|^{1/2(n-p-1)} \\ \cdot |I-L|^{1/2(m-p-1)} \int_{T>0} \text{etr}(-1/2 \Sigma^{-1} T) |T|^{1/2(m+n-p-1)} \\ \cdot {}_0F_1(1/2 n; 1/2 \Omega \Sigma^{-1} T^{1/2} L T^{1/2}) dT, \quad 0 < L < I.$$

$$(3.2) \quad (ii) \quad f_{\mathcal{L}}(L) = \text{etr}(-1/2 \Omega) {}_1F_1(1/2(m+n); 1/2 n; 1/2 \Omega, L).$$

$$\cdot \left[\frac{1}{\beta_p(1/2 m, 1/2 n)} |L|^{1/2(n-p-1)} |I-L|^{1/2(m-p-1)} \right]$$

$$\text{where } \beta_p(\frac{1}{2}m, \frac{1}{2}n) = \frac{\Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n)}{\Gamma_p(\frac{1}{2}(m+n))}$$

(Note: The expression in square brackets is the central Beta type 1 density function.)

$$(iii) \quad 1 > \theta_1 > \dots > \theta_p > 0 \text{ are the roots of } |B - \theta(A+B)| = 0.$$

Then the joint density of the θ_i is

$$h(\theta_1, \dots, \theta_p) = \text{etr}(-\frac{1}{2}\Omega) {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega, D_\theta).$$

$$\cdot \frac{\pi^{\frac{1}{2}p^2} \Gamma_p(\frac{1}{2}(m+n))}{\Gamma_p(\frac{1}{2}p) \Gamma_p(\frac{1}{2}m) \Gamma_p(\frac{1}{2}n)} \prod_{i=1}^p \theta_i^{\frac{1}{2}(n-p-1)} \\ \cdot (1-\theta_i)^{\frac{1}{2}(m-p-1)} \prod_{i < j}^p (\theta_i - \theta_j)$$

Proof:

Part (i) is proved using transformation of variables (see De Waal (1968)). Part (ii) is easily proved using (2.3) and (3.1) and part (iii) follows immediately from part (ii) by virtue of Theorem 2.2, seeing that the symmetrised density is in the form (2.4) required by the theorem, and Theorem 2.1.

Similar results are easily obtained for the following Beta matrix variates; the same definitions for matrices A and B apply as before.

$$(3.3) \quad (i) \quad M\beta_{1a} \text{ matrix } G = (A+B)^{-\frac{1}{2}} A (A+B)^{-\frac{1}{2}} \quad (I = I-L)$$

$$(ii) \quad M\beta_{2a} \text{ matrix } V = B^{-\frac{1}{2}} A B^{-\frac{1}{2}}$$

$$(iii) \quad M\beta_{2b} \text{ matrix } Z = B^{\frac{1}{2}} A^{-1} B^{\frac{1}{2}} \quad (= V^{-1}) *$$

Again for the Beta type 2 variates we find that the symmetrised noncentral density is in the rather satisfying form of a weighted central density. For example, the symmetrised density of V defined above is

$$f_s(V) = \text{etr}(-\frac{1}{2}\Omega) {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}n; \frac{1}{2}\Omega, (I+V)^{-1}).$$

$$\cdot \left[\frac{1}{\beta_p(\frac{1}{2}m, \frac{1}{2}n)} |V|^{\frac{1}{2}(m-p-1)} |I+V|^{-\frac{1}{2}(m+n)} \right], \quad V > 0$$

* This definition of the $M\beta_{2b}$ matrix variate avoids problems in integration which occur when deriving the distribution of the more correctly defined variate $A^{-\frac{1}{2}}BA^{-\frac{1}{2}}$; in any case the properties in which we are interested do not depend on this definition.

where the expression in square brackets is the central Beta type 2 density function.

4. Noncentral distribution of the characteristic roots of the generalised sample multiple correlation matrix

A further illustration of the use of symmetrised distributions is given here. Referring to the hypothesis example (i) in the introduction, the Beta matrix from which the test criteria for this hypothesis are derived is the generalised sample multiple correlation matrix as defined by Khatri (1964) and Troskie (1969),

$$R = A_{11}^{-1/2} A_{12} A_{22}^{-1} A_{21} A_{11}^{-1/2}$$

where the $A_{ij}; p_i \times p_j$ matrices are the usual variance and covariance matrices between the two sets of variables for $N = n+1$ observations. The corresponding population matrices are

$$P = \Sigma_{11}^{-1/2} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1/2}$$

R is distributed according to a multivariate Beta type 1b distribution conditional on the second set of variables being fixed (in fact conditional on fixed A_{22}). To obtain ultimately the unconditional distribution of the characteristic roots of R we pass immediately into the symmetrised forms. Firstly the symmetrised conditional density of R may be written down immediately from (3.2) with the necessary parameter substitutions ' m ' = $n-p_2$, ' n ' = p_2 , ' p ' = p_1 . By multiplying this density by the Wishart density of A_{22} (with parameter matrix Σ_{22} and n degrees of freedom) and then integrating with respect to A_{22} , the symmetrised unconditional density of R is obtained:

$$f_s(R) = |I-P|^{1/2 n} {}_2F_1\left(\frac{1}{2}n, \frac{1}{2}n; \frac{1}{2}p_2; P, R\right) \cdot \int \frac{1}{\beta_{p_1}(\frac{1}{2}(n-p_2), \frac{1}{2}p_2)} |R|^{1/2(p_2-p_1-1)} |I-R|^{1/2(n-p_2-p_1-1)} dR$$

Once again $f_s(R)$ is in the form of a central Beta density multiplied by a weighting function, and the distribution of the roots of R can be obtained as an immediate consequence of Theorem 2.2

ACKNOWLEDGEMENTS

The author wishes to express his gratitude to Prof. C.G. Troskie and Dr J.J.J. Roux for their valuable suggestions.

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Manuscript received 10 May 1973.